TRDP

Train Real Time Data Protocol

3.0.0.0 Conformance Test Report

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DOCUMENT SUMMARY SHEET

This document contains the TRDP test report

Participants		
Name and Surname	Organisation	Role
Peter Liesner	NewTec	Participant

History			
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1. Introduction

1.1. Purpose

This document describes tests verifying the conformance of a TRDP implementation.

1.2. Intended Audience

This document is intended to be used for verification of the TCNOpen TRDP implementation.

1.3. References/Related Documents

Reference	Number	Title
[Wire]	IEC61375-2-3	TRDP Protocol (Annex A)
[Req]	TCN-TRDP1-D-BOM-003	TRDP System Requirement Specification
[TestRep]	TCN-TRDP1-D-BOM-032	TRDP Conformance Test Report

Table 1: References

1.4. Abbreviations and Definitions

Abbreviation	Definition

Table 2: Abbreviations and Definitions

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2. Conformance Tests

The following tests verify the conformance of the TCNOpen TRDP implementation for a specified TRDP Version 2.1.0.0. For testing the programs “test/pdpatterns/trdp-pd-test.c-fast.c” and “test/mdpatterns/trdp-md-test-fast.c” delivered with this TRDP version shall be used as well as trdp-pd-test.c and trdp-md-test.c

For documentation of the test results [TestRep] shall be used as template.

2.1. Process Data

PD tests verify the exchange of process data between two devices A and B. All in [wire] defined PD patterns are tested.

All the test cases run continuously in parallel during the test session.

2.1.1. Testconfiguration:

IP address device A: 10.0.1.26

IP address device B: 10.0.1.30

Multicast address: 239.0.1.1

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2.1.2. PD1: Windows/TCNOpen - Linux/TCNOpen:

Pattern	Destination	Direction	Data Size in Bytes	Period in ms	Result
PUSH	unicast	A->B, B->A	256	100	OK
				250	OK
			1432	100	OK
				250	OK
	multicast	A->B, B->A	256	100	OK
				250	OK
			1432	100	OK
				250	OK
PULL	unicast / unicast	A->B->A, B->A->B	256	500	OK
			1432	500	OK
	multicast / multicast	A->B->A, B->A->B	256	500	OK
			1432	500	OK

Table 3: PD Test Patterns(pdtest-fast-windows)

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2.1.3. PD1: Linux/TCNOpen - Windows/TCNOpen:

Pattern	Destination	Direction	Data Size in Bytes	Period in ms	Result
PUSH	unicast	A->B, B->A	256	100	OK
				250	OK
			1432	100	OK
				250	OK
	multicast	A->B, B->A	256	100	OK
				250	OK
			1432	100	OK
				250	OK
PULL	unicast / unicast	A->B->A, B->A->B	256	500	OK
			1432	500	OK
	multicast / multicast	A->B->A, B->A->B	256	500	OK
			1432	500	OK

Table 4: PD Test Patterns(pdtest-fast-linux)

All the test cases run on two target platforms - Linux and Windows.

Following table summarizes the test targets used for the specific tests:

	Device A	Device B
PD1	Windows/TCNOpen	Linux/TCNOpen

Table 5: PD Test Targets

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2.2. Message Data

MD tests verify the exchange of message data between two devices A and B.

All defined MD patterns are tested on both supported transmission protocols TCP and UDP.

2.2.1. Testconfiguration:

IP address device A: 10.0.1.26

IP address device B: 10.0.1.30

Multicast address: 239.0.1.1

Following table defines the test cases performed on the two platforms and implementations:

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2.2.2. MD1: Windows/TCNOpen - Linux/TCNOpen

Protocol	Pattern	Destination	Re-plies	Direction	Data Size in Bytes	Result
UDP	notify	unicast	0	A->B	64	OK
					32768	OK
	request/reply	unicast/unicast	1	A->B->A	64	OK
					32768	OK
	request/reply/confirm	unicast/unicast	1	A->B->A->B	64	OK
					32768	OK
	notify	multicast	0	A->B	64	OK
					32768	OK
	request/reply	multicast/unicast	1	A->B->A	64	OK
					32768	OK
	request/reply/confirm	multicast/unicast	1	A->B->A->B	64	OK
					32768	OK
	request/reply	multicast/unicast	?	A->B->A	64	OK
					32768	OK
	request/reply/confirm	multicast/unicast	?	A->B->A->B	64	OK
					32768	OK
TCP	notify	unicast	0	A->B	64	OK
					32768	OK
	request/reply	unicast	1	A->B->A	64	OK
					32768	OK
	request/reply/confirm	unicast	1	A->B->A->B	64	OK
					32768	OK

Table 6: MD Test Patterns

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2.2.3. MD2: Linux/TCNOpen - Windows/TCNOpen

Protocol	Pattern	Destination	Re-plies	Direction	Data Size in Bytes	Result
UDP	notify	unicast	0	A->B	64	OK
					32768	OK
	request/reply	unicast/unicast	1	A->B->A	64	OK
					32768	OK
	request/reply/confirm	unicast/unicast	1	A->B->A->B	64	OK
					32768	OK
	notify	multicast	0	A->B	64	OK
					32768	OK
	request/reply	multicast/unicast	1	A->B->A	64	OK
					32768	OK
	request/reply/confirm	multicast/unicast	1	A->B->A->B	64	OK
					32768	OK
	request/reply	multicast/unicast	?	A->B->A	64	OK
					32768	OK
	request/reply/confirm	multicast/unicast	?	A->B->A->B	64	OK
					32768	OK
TCP	notify	unicast	0	A->B	64	OK
					32768	OK
	request/reply	unicast	1	A->B->A	64	OK
					32768	OK
	request/reply/confirm	unicast	1	A->B->A->B	64	OK
					32768	OK

Table 7: MD Test Patterns

All the test cases run on two target platforms - Linux and Windows.

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Following table summarizes the tests performed on the two platforms and implementations:

	Device A	**Device B**
MD1	Windows/TCNOpen	Linux/TCNOpen
MD2	Linux/TCNOpen	Windows/TCNOpen

Table 8: MD Test Targets

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